# 数据保护和云
## ONTAP 9

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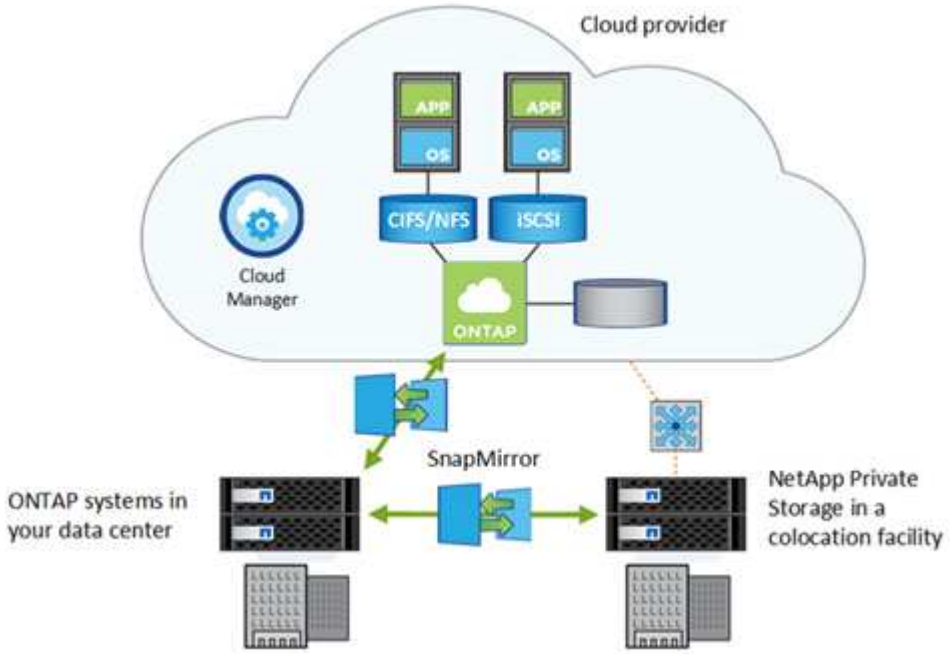
# 数据保护和云

## 数据保护和云概述

数据保护通常是客户开始云之旅时首先尝试的事情。保护可以像密钥数据的异步复制一样简单，也可以像整个热备份站点一样复杂。数据保护主要基于熟悉的 NetApp SnapMirror 技术。

## 数据复制

SnapMirror 技术可使用 ONTAP Snapshot 副本在内部和云安装之间保持数据同步。SnapMirror 执行块级增量数据传输，以确保仅将更改的数据发送到目标副本。




同样，您也可以使用 SnapMirror 存储关系为在 Cloud Volumes ONTAP 系统上创建的本地 Snapshot 副本创建数据归档。

NetApp Cloud Backup 提供无缝且经济高效的备份和还原功能，用于保护数据并将其归档到云中的对象存储。Cloud Backup 既可用于基于云的数据，也可用于内部数据。

"使用 Cloud Volumes ONTAP 在云中设置灾难恢复"

"使用 Cloud Volumes ONTAP 和 SnapMirror 进行高效数据复制"

"使用命令行界面保护 ONTAP 数据"

"NetApp Cloud Backup"

# 高可用性

在内部数据中心中，物理节点配置为高可用性（HA）对，以实现容错和无中断运行。如果某个节点发生故障或需要关闭某个节点进行日常维护，则其配对节点将接管其存储并继续从该节点提供数据。

在云环境中，您可以创建一个 Cloud Volumes ONTAP 实例 HA 对，以实现与内部 HA 对相同的容错和无中断操作。这些恢复目标适用于云 HA 对：

- 恢复点目标（RPO）为 0 秒。您的数据在传输过程中不会丢失数据。
- 恢复时间目标（RTO）为 60 秒。如果发生中断、数据应在 60 秒或更短的时间内可用。

每个云提供商都提供自己的 HA 架构和配置选项。对于 Cloud Volumes Service，服务级别协议保证高可用性。

"AWS 中的高可用性对"

"Azure 中的高可用性对"

# 空闲数据加密

ONTAP 使用与保护内部数据相同的加密技术保护云中的数据。

Cloud Volumes ONTAP 通过外部密钥管理器支持 NetApp 卷加密（NVE）和 NetApp 聚合加密（NAE）。

Cloud Volumes ONTAP 还支持以下加密技术：

- AWS 密钥管理服务
- Azure 存储服务加密
- Google Cloud Platform 默认加密

使用 Azure NetApp Files 和适用于 Google Cloud 的 NetApp Cloud Volumes Service 时，数据始终会在空闲时加密。

"在 Cloud Volumes ONTAP 中对空闲数据进行加密"

"NetApp 卷加密和 NetApp 聚合加密"

"使用 NetApp 加密解决方案对 Cloud Volumes ONTAP 中的卷进行加密"

# 防病毒保护

您可能会在内部使用集成的防病毒功能来防止数据受到病毒或其他恶意代码的侵害。使用 Cloud Volumes ONTAP 时，云中也提供同样的防病毒保护。

安装在本地服务器上的 ONTAP 防病毒连接器可处理存储系统与防病毒软件之间的通信。对于 Cloud Volumes ONTAP，您可以在与 ONTAP 位于同一云中的虚拟机上安装 Antivirus Connector。

"防病毒配置"

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